



1) You are building ----- was not required?

Ans Choice : ESP32

Reasons :

- built in Wi-Fi and Bluetooth
- can send data directly to a mobile app / cloud
- faster processor (240 MHz vs Arduino UNO's 16 MHz)
- More memory and GPIO pins
- suitable for IoT applications

WORKING

- 1) soil moisture sensor measures moisture
 - 2) ESP32 reads sensor data
 - 3) if moisture is below threshold, ESP32 turns ON pump through relay
 - 4) Data sent to mobile application every 5 seconds.
- of internet connectivity



is not required.

Arduino Uno can be used because:

- simpler
- lower cost
- enough for local monitoring and pump control.

2. A smart factory contains 500 sensors --- choose and why?

Ans - Comparison

PROTOCOL	RANGE	POWER
Wi-Fi	50-100 m	High
Bluetooth	10-100 m	Low
Lo Ra	2-15 km	Very Low
MQTT	Application protocol	Depends on network.



Date ___/___/___

Protocol Speed
WiFi Very High

Bluetooth Medium

LoRa Low

MQTT Efficient

Best Combination

LoRa + MQTT

Why?

- LoRa connects 500 sensors over 2km.
- Gateway collects sensor data.
- MQTT sends data efficiently to cloud/server.
- low power consumption and large coverage.



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3. A temp. sensor - - - - ADC or software

Ans Step by step troubleshooting

1) check sensor

- replace with known working sensor
- compare readings

2) check wiring

- loose wires
- broken connection
- wrong pin connections

3) check power supply

- measure voltage using multi-meter
- ensure stable power

4) Check ADC

- read raw ADC values
- compare with expected values



5) check software

- verify calibration formula
- print raw data using serial monitor
- remove scaling errors -

CONCLUSION →

by testing one component at a time, the faulty section can be identified.

4) An ESP32-based weather -- battery life?

Ans Hardware modifications

- use low-power sensors
- use lithium battery
- add solar charging if possible
- use efficient voltage regulator.



Software modifications →

- deep sleep mode
- wake every 10-15 minutes
- Turn Wi-Fi ON only during transmission
- reduce OLED display usage
- compress transmitted data

Result :

power consumption decreases significantly, extending battery life.

5. an automatic braking ---- and choose and why?

Ans Best choice

CAN bus (Controller area network)

Reasons : →

- Very low latency
- high reliability



- error detection mechanisms
- widely used in automobiles

ALTERNATIVE →

automotive ethernet for advanced systems.

6. You need to measure --- circle length 'a' and height 'b'

Ans **PLACEMENT**

place sensor:

- at center of room.
- mid-height (approximately 5ft)
- away from windows, doors, fans, AC vents, and direct sunlight.

WHY?

- represents average air-temperature.

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→ minimizes local hot / cold effects.

POOR PLACEMENT EFFECTS →

- near window → higher temperature
- near AC → lower temperature
- Near ceiling → hotter air
- near door → fluctuating readings

7. under what situations --- a digital one?

Ans situations

- 1) CONTINUOUS MEASUREMENTS
analog output changes smoothly.
- 2) HIGHER RESOLUTION REQUIREMENTS
not limited by digital quantization
- 3) FAST SIGNAL CHANGES
no conversion delay.
- 4) LOWER COST APPLICATIONS
many analog sensors are cheaper.

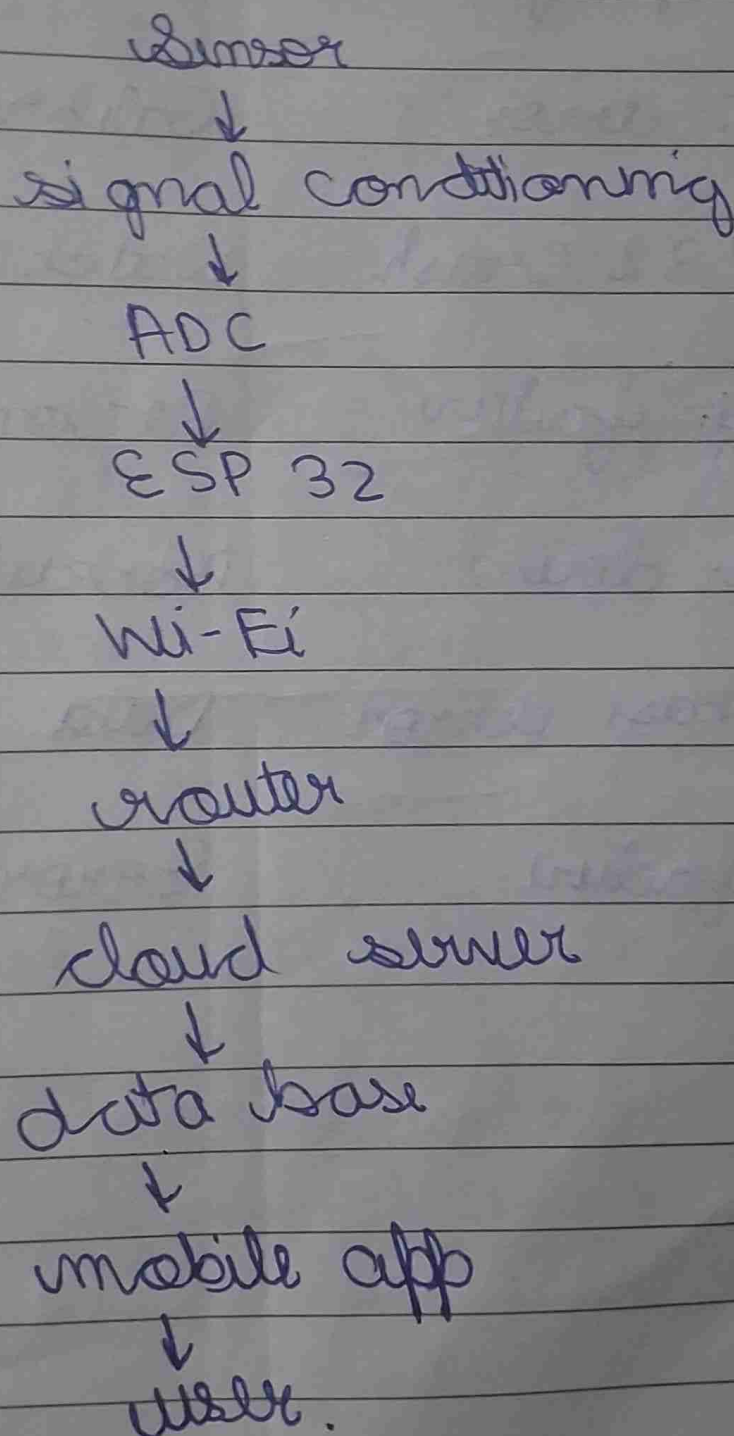


EXAMPLE

Thermistor measuring temp -
excited continuously.

8 Draw the complete --- one solution
on for each.

Ans



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FAILURE POINTS AND SOLUTIONS

failure	Solution
sensor failure	redundant sensor
wiring fault	shielded cables
ADC error	Calibration
ESP32 Crash	Watch dog Timer
Wi-fi failure	Reconnect logic
Server down	Backup server
Data base error	Data replication
App failure	Error handling



- Q) If an attacker - - - do not cite far-fetched ideas

Ans POSSIBLE ATTACKS

- 1) Unauthorized access
- 2) Password attacks
- 3) Data interception
- 4) Malware injection
- 5) Denial of Service (DoS)

Security Improvements:

- ① Strong password
Unique credentials
- ② WPA3 encryption
Secure Wi-Fi communication
- ③ HTTPS / TLS
encrypted data transmission
- ④ Secure firmware updates
Signed firmware only -



- ⑤ firewall and access control restrict unauthorized devices
- ⑥ disable unused ports reduce attack surface.
- ⑩ Your ESP32 must -- you solve them?

Ans POSSIBLE CONFLICTS

Hardware

- GPIO pin conflicts
- power shortage
- I2C address conflicts

Software

- Timing conflicts
- blocking code
- memory issues

SOLUTIONS

- proper pin planning
- separate power supply.



- non-blocking programming
- FreeRTOS tasks
- I²C address management.

1) An ESP32 uploads - - - - - permanently lost

Ans Strategy

- ① store data in
 - EEPROM
 - SD card
 - Flash Memory
- ② add timestamp
- ③ retry connection periodically
- ④ upload stored data after internet returns -
- ⑤ confirm successful upload before deleting local copy.



BENEFIT

no important data is lost.

12. your IOT device -- long term reliability?

Ans Hardware Improvements

- industrial-grade components
- surge protection
- proper heat dissipation
- water proof enclosure
- quality power supply

Software Improvements

- watchdog timer
- error recovery
- memory leak prevention
- OTA updates
- data integrity checks

result:

improved long term reliability



15 Can an IOT system be "intelligent"?

Ans. yes, an IOT system can be considered smart even without artificial intelligence AI

- smart IOT system can sense its environment, process information and perform actions automatically based on predefined rules.
- for eg → a smart irrigation system can automatically switch on a water pump when soil moisture falls below a certain threshold.
- Similarly, a motion sensor can automatically turn on lights when a person enters a room.
- These systems make decisions based on fixed programming rather than learning from data.

DIFFERENCE

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→ smart system

- operates using pre-defined rules and conditions
- performs automatic actions without human intervention
- does not learn from past experiences
- behaviour remains fixed unless reprogrammed.

→ intelligent system: →

- uses AI and machine learning techniques
- learns from historical data and improves performance over time
- can identify patterns and make predictions
- adapts its behaviour according to changing conditions.

EXAMPLE →

→ A smart Thermostat turns on cooling when temp. exceeds a set value.



→ an intelligent thermostat learns user preferences and predicts when cooling will be needed.

CONCLUSION →

all intelligent systems are smart, but not all smart systems are intelligent.

AI enhances IOT systems by making them adaptive, predictive and capable of learning from experience.

14. Compare polling ---- IOT device?

Ans. POLLING

- The micro-controller continuously checks the status of sensors and devices.
- The CPU remains busy checking whether an event has occurred.
- It is simple to implement but consumes more processing power.

ADVANTAGES →

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- easy to understand and program.
- suitable for simple applications.

DISADVANTAGES

- wastes CPU Time
- higher power consumption
- slower response to urgent events

INTERRUPT-BASED PROGRAMMING

- devices send an interrupt signal when an event occurs
- The CPU responds immediately and executes the required task.
- More efficient than polling

ADVANTAGES

fast response time
better CPU utilization
lower power consumption



applications of interrupts

- fire alarm systems
- motion detection
- security systems
- push-button control
- vehicle safety systems

CONCLUSION

- polling is suitable for simple systems
- interrupts are preferred in RT applications requiring quick response and energy efficiency.

15. Energy engineering - - were justified

Energy engineering design in -
values trade-off b/w cost
performance power consump-
tion and reliability -



① cost vs accuracy

- low-cost sensors were used instead of industrial-grade sensors
- This reduced project cost but slightly affected accuracy
- The trade off was acceptable because accuracy was sufficient for the application

② Battery life vs update frequency

- data was transmitted at intervals instead of continuously
- This increased battery life and reduced power consumption
- a small delay in updates was considered acceptable

③ simplicity vs advanced features

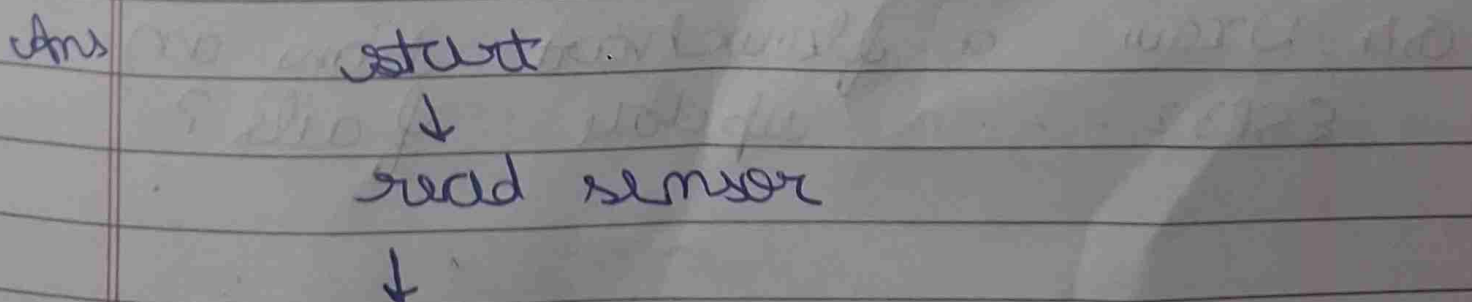


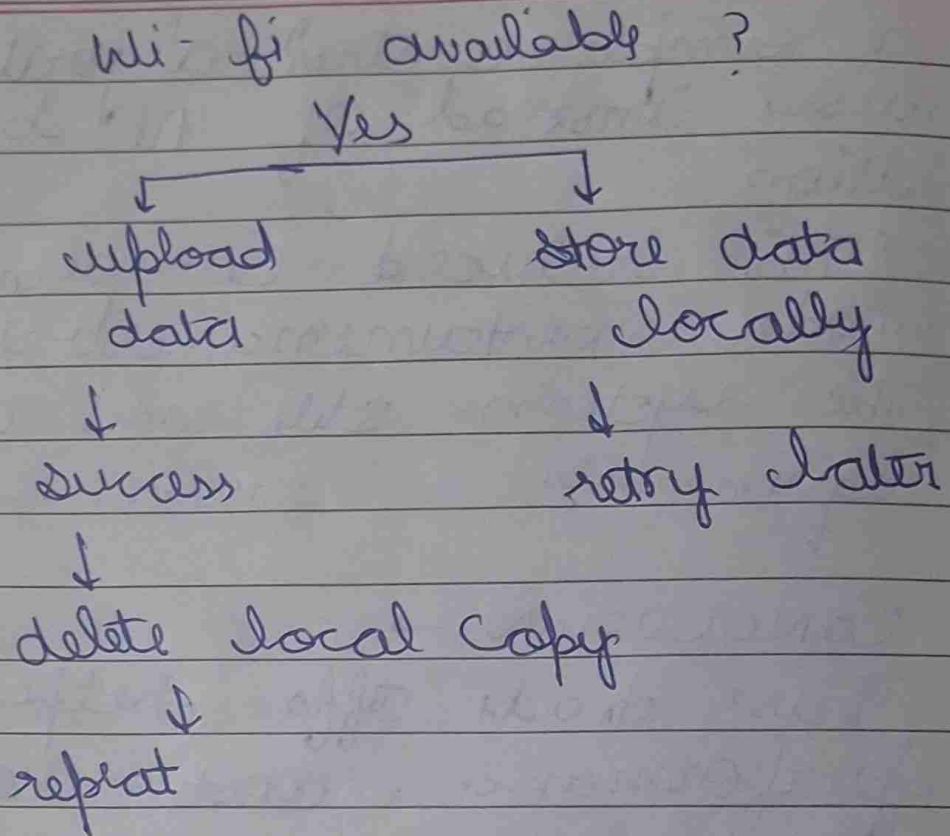
- a simple control algorithm was used instead of AI based automation
- This reduced complexity and made maintenance easier
- The system still met all project requirements

CONCLUSION →

- These trade offs helped balancing performance, cost and efficiency
- The decisions were justified because they met the project's objectives while keeping the system practical & economical.

Q16. create a flowchart used in system.





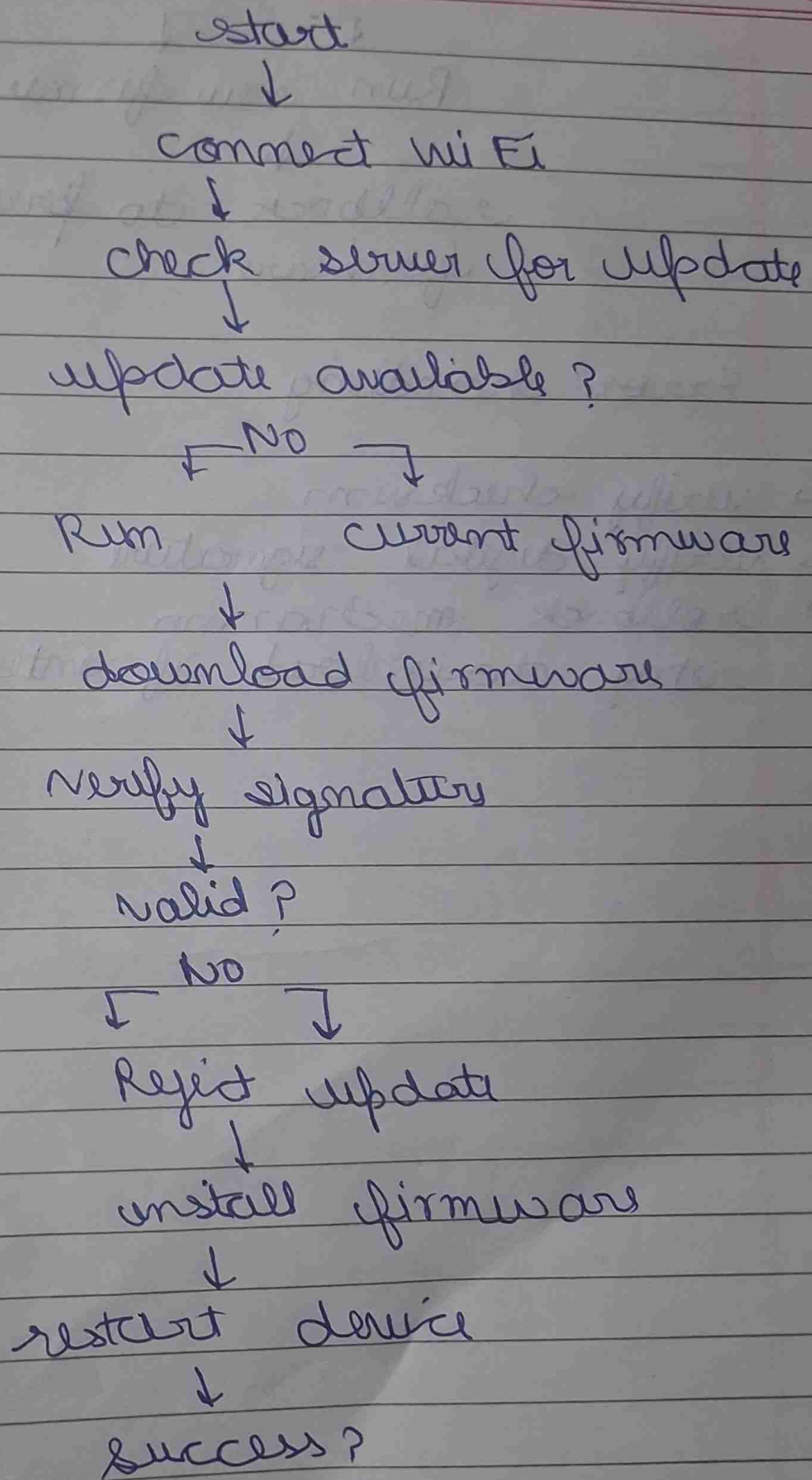
Component used

- ESP32
- sensor
- flash/SD storage
- Wi Fi module
- MQTT/ HTTP protocol
- cloud server

Q7 Draw a flowchart how an ESP32 - - - update fails ?



Ans





Yes
↓

Run new firmware

↓

rollback to previous
firmware

Error handling

- verify checksum
- verify digital signature
- rollback mechanism
- retry download if interrupted

↓

multiple files

↓

9 files

out

↓

↓

to file - to file

↓

encrypt, decrypt

↓

write - backup

↓

9 files